Female Labour Force Participation in India

A State-Level Analysis Using PLFS 2021–22

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1 Plot settings

```r
main_colour <- "#19324A"
point_colour <- "#356A7D"
accent_colour <- "#2A7F7F"
highlight_colour <- "#C47A44"

theme_set(
  theme_minimal(base_size = 11) +
    theme(
      plot.title = element_text(
        face = "bold",
        size = 13,
        colour = main_colour
      ),
      plot.subtitle = element_text(
        size = 10
      ),
      axis.title = element_text(
        colour = "#333333"
      ),
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(
        colour = "#E5E5E5"
      ),
      plot.caption = element_text(
        size = 9,
        colour = "#666666"
      )
    )
)
```

2 Import cleaned working dataset

```r
df <- read_excel("main_female_lfpr_dataset_cleaned.xlsx")
```

3 Inspect imported data

```r
# Data preparation

# Helper function to safely convert imported values to numeric

clean_numeric <- function(x) {
  if (is.numeric(x)) {
```

```
    return(x)
  }

  x <- trimws(as.character(x))
  x[x %in% c("", "NA", "N/A", "-", "-")] <- NA

  as.numeric(gsub("[^0-9.-]", "", x))
}

# Create analytical variables

df_model <- df %>%
  filter(state_ut != "Andaman & N. Island") %>%
  mutate(
    female_lfpr = clean_numeric(`female_lfpr_%`),
    secondary_above = clean_numeric(`female_secondary_above_%`),
    nsdp_numeric = clean_numeric(nsdp_2021_22),
    log_nsdp = ifelse(
      !is.na(nsdp_numeric) & nsdp_numeric > 0,
      log(nsdp_numeric),
      NA_real_
    ),
    urbanisation = clean_numeric(`urbanisation_2011_%`),
    female_non_agri_share = clean_numeric(`female_non_agri_share_%`),
    married_before_18 = clean_numeric(`women_married_before_18_%`),
    tfr = clean_numeric(`TFR %`)
  )

# Analytical sample without requiring NSDP

df_analysis <- df_model %>%
  drop_na(
    female_lfpr,
    secondary_above,
    urbanisation,
    female_non_agri_share,
    married_before_18,
    tfr
  )

# Analytical sample for models requiring NSDP

df_nsdp <- df_analysis %>%
  drop_na(log_nsdp)

# Check variable classes
```

```r
sapply(
  df_analysis %>%
    select(
      female_lfpr,
      secondary_above,
      urbanisation,
      female_non_agri_share,
      married_before_18,
      tfr
    ),
  class
)
```

```
##           female_lfpr       secondary_above          urbanisation
##             "numeric"             "numeric"             "numeric"
## female_non_agri_share     married_before_18                   tfr
##             "numeric"             "numeric"             "numeric"
```

```r
# Check sample sizes

dim(df_analysis)
```

```
## [1] 33 16
```

```r
dim(df_nsdp)
```

```
## [1] 32 16
```

```r
df %>%
  select(state_ut, nsdp_2021_22)
```

```
## # A tibble: 34 x 2
##    state_ut                           nsdp_2021_22
##    <chr>                              <chr>
##  1 Andhra Pradesh                     61838660
##  2 Arunachal Pradesh                  1616693
##  3 Assam                              22432223
##  4 Bihar                              34358255
##  5 Chandigarh                         2700338
##  6 Chhattisgarh                       23264575
##  7 Dadra & Nagar Haveli & Daman & Diu NA
##  8 Delhi                              49885230
##  9 Goa                                4604014
## 10 Gujarat                            119856386
## # i 24 more rows
```

3.1 Final Analytical Sample

```
# Create the main analytical sample
# All variables except log(NSDP) are available for all 34 states.

df_analysis <- df_model %>%
  drop_na(
    female_lfpr,
    secondary_above,
    urbanisation,
    female_non_agri_share,
    married_before_18,
    tfr
  )

# Create the NSDP analytical sample
# Two states have missing NSDP values.

df_nsdp <- df_analysis %>%
  drop_na(log_nsdp)

# Check sample sizes

cat("General analytical sample:", nrow(df_analysis), "states\n")
```

```
## General analytical sample: 33 states
```

```
cat("NSDP analytical sample:", nrow(df_nsdp), "states\n")
```

```
## NSDP analytical sample: 32 states
```

```
# Variable-level availability

sample_sizes <- data.frame(
  Variable = c(
    "Female LFPR",
    "Secondary education and above",
    "Log NSDP",
    "Urbanisation",
    "Female non-agricultural employment share",
    "Women married before age 18",
    "Total fertility rate"
  ),
  N = c(
    sum(!is.na(df_analysis$female_lfpr)),
```

```
    sum(!is.na(df_analysis$secondary_above)),
    sum(!is.na(df_analysis$log_nsdp)),
    sum(!is.na(df_analysis$urbanisation)),
    sum(!is.na(df_analysis$female_non_agri_share)),
    sum(!is.na(df_analysis$married_before_18)),
    sum(!is.na(df_analysis$tfr))
  )
)

knitr::kable(
  sample_sizes,
  caption = "Available observations by analytical variable"
)
```

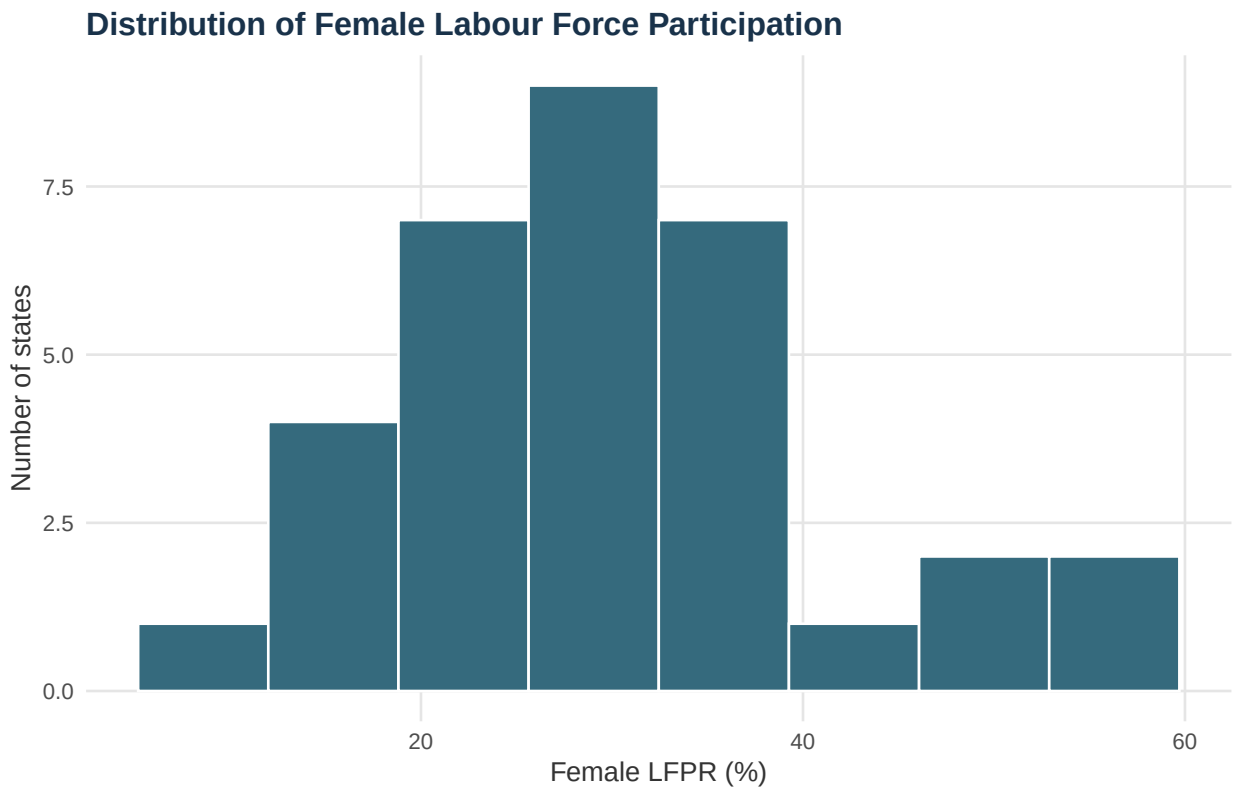
Table 1: Available observations by analytical variable

Variable	N
Female LFPR	33
Secondary education and above	33
Log NSDP	32
Urbanisation	33
Female non-agricultural employment share	33
Women married before age 18	33
Total fertility rate	33

4 Descriptive Analysis

4.1 Distribution of Female LFPR

```
ggplot(df_analysis, aes(x = female_lfpr)) +
  geom_histogram(
    bins = 8,
    fill = point_colour,
    color = "white"
  ) +
  labs(
    title = "Distribution of Female Labour Force Participation",
    x = "Female LFPR (%)",
    y = "Number of states"
  )
```



4.2 Descriptive Statistics

```
descriptive_statistics <- df_analysis %>%
  summarise(
    across(
      c(
        female_lfpr,
        secondary_above,
        log_nsdp,
        urbanisation,
        female_non_agri_share,
        married_before_18,
        tfr
      ),
      list(
        Mean = ~mean(.x, na.rm = TRUE),
        SD = ~sd(.x, na.rm = TRUE),
        Minimum = ~min(.x, na.rm = TRUE),
        Maximum = ~max(.x, na.rm = TRUE)
      )
    )
  )
```

```
descriptive_statistics
```

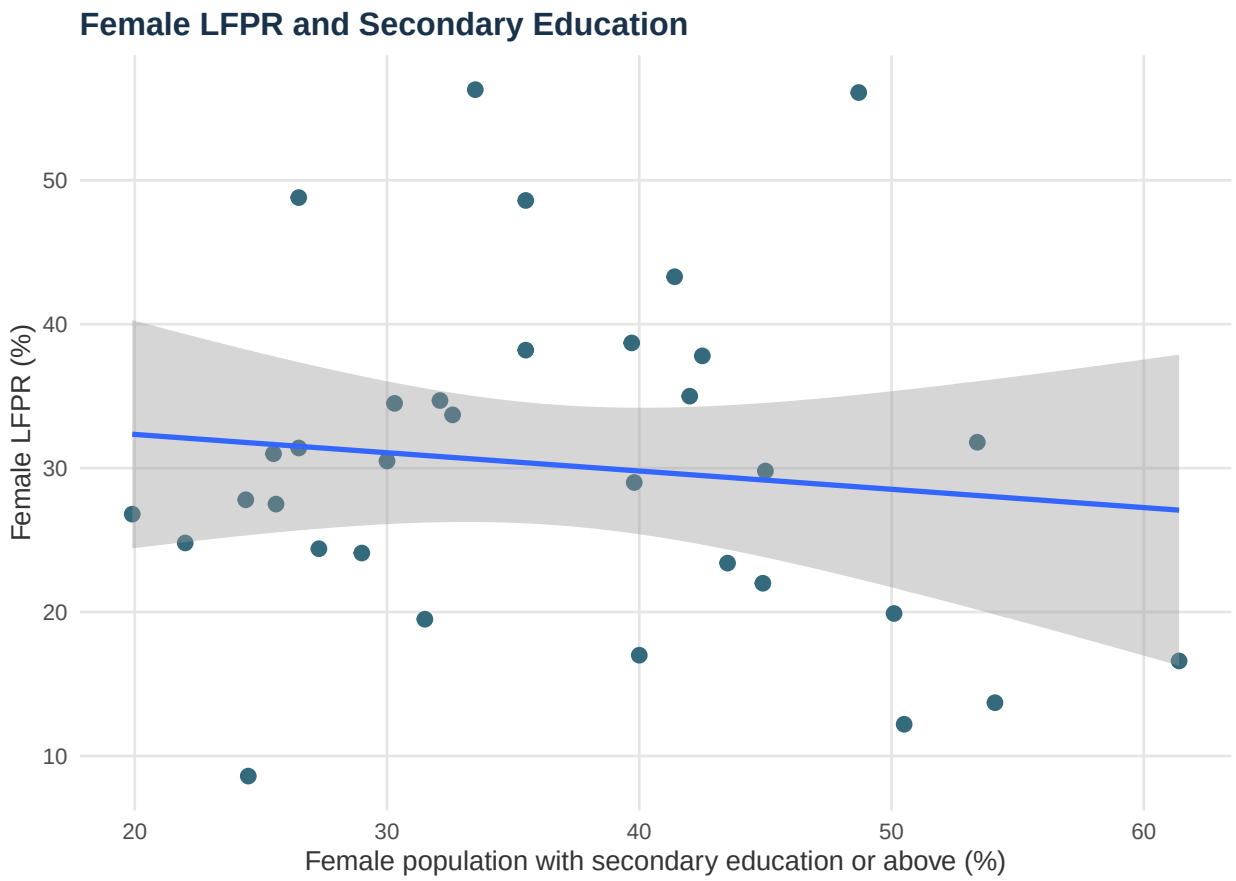
```
## # A tibble: 1 x 28
##   female_lfpr_Mean female_lfpr_SD female_lfpr_Minimum female_lfpr_Maximum
##              <dbl>          <dbl>                <dbl>                <dbl>
## 1             30.2           11.7                  8.6                 56.3
## # i 24 more variables: secondary_above_Mean <dbl>, secondary_above_SD <dbl>,
## #   secondary_above_Minimum <dbl>, secondary_above_Maximum <dbl>,
## #   log_nsdp_Mean <dbl>, log_nsdp_SD <dbl>, log_nsdp_Minimum <dbl>,
## #   log_nsdp_Maximum <dbl>, urbanisation_Mean <dbl>, urbanisation_SD <dbl>,
## #   urbanisation_Minimum <dbl>, urbanisation_Maximum <dbl>,
## #   female_non_agri_share_Mean <dbl>, female_non_agri_share_SD <dbl>,
## #   female_non_agri_share_Minimum <dbl>, ...
```

5 Bivariate Analysis

5.1 Female LFPR and Secondary Education

```
ggplot(
  df_analysis,
  aes(x = secondary_above, y = female_lfpr)
) +
  geom_point(
    size = 2.5,
    color = point_colour
  ) +
  geom_smooth(
    method = "lm",
    se = TRUE
  ) +
  labs(
    title = "Female LFPR and Secondary Education",
    x = "Female population with secondary education or above (%)",
    y = "Female LFPR (%)"
  )
```

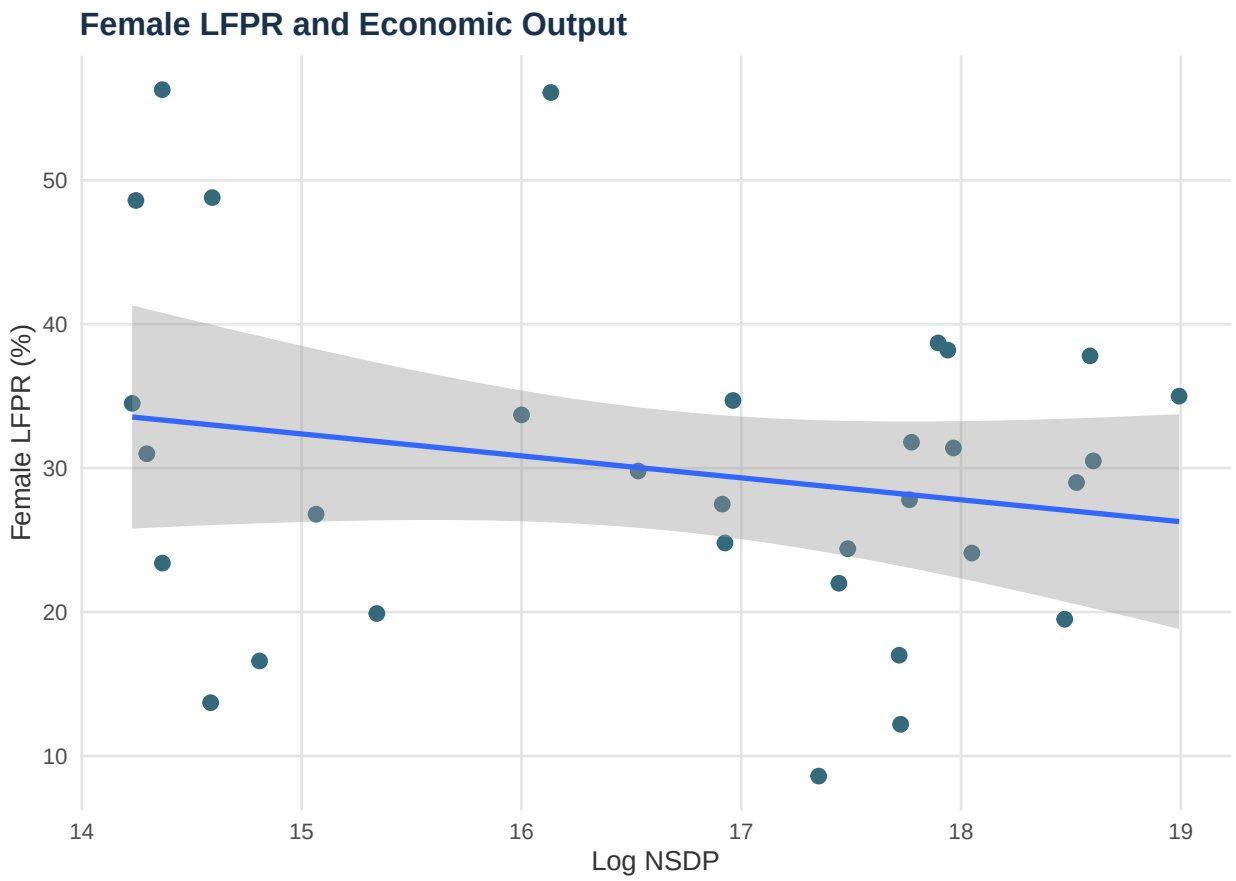
```
## `geom_smooth()` using formula = 'y ~ x'
```



5.2 Female LFPR and Economic Output

```
ggplot(
  df_nsdp,
  aes(x = log_nsdp, y = female_lfpr)
) +
  geom_point(
    size = 2.5,
    color = point_colour
  ) +
  geom_smooth(
    method = "lm",
    se = TRUE
  ) +
  labs(
    title = "Female LFPR and Economic Output",
    x = "Log NSDP",
    y = "Female LFPR (%)"
  )
```

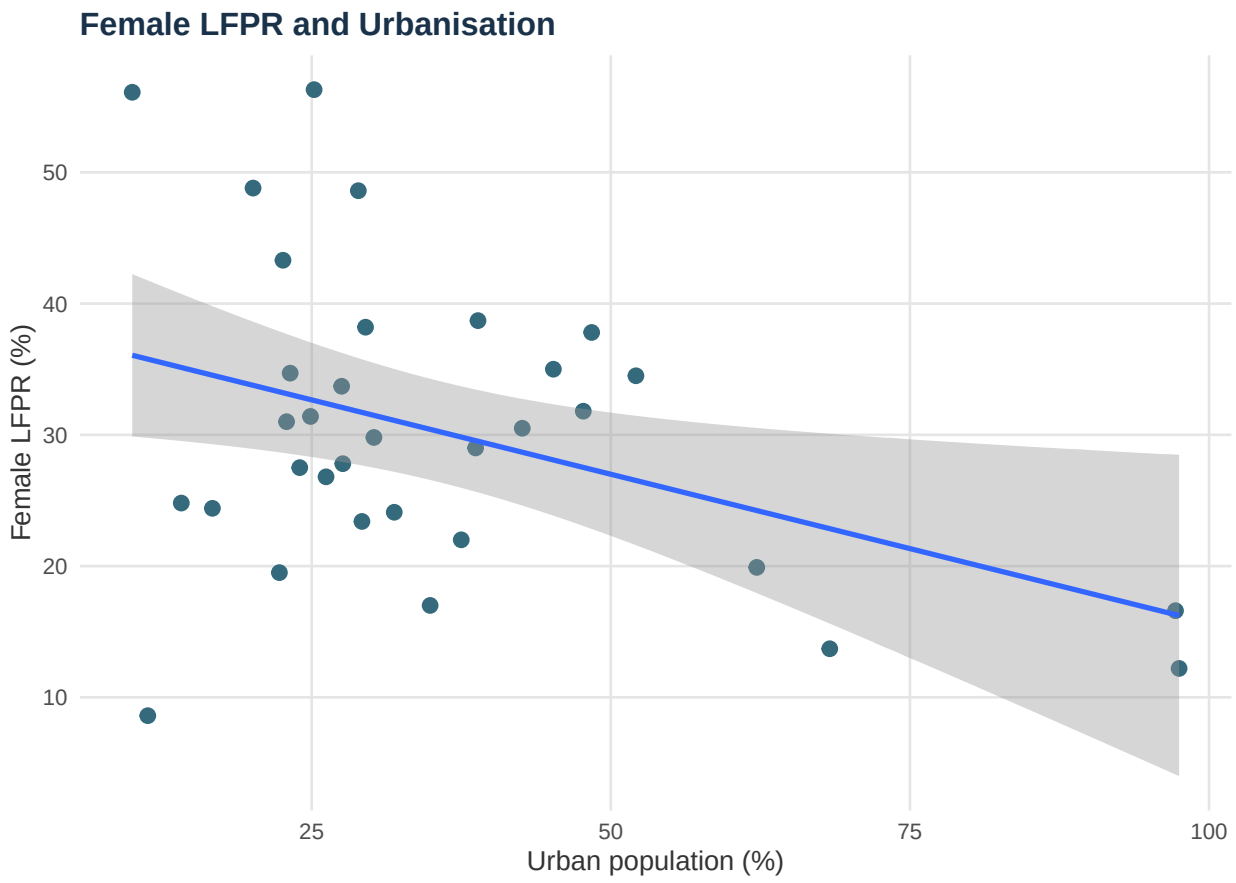
```
## `geom_smooth()` using formula = 'y ~ x'
```



5.3 Female LFPR and Urbanisation

```
ggplot(
  df_analysis,
  aes(x = urbanisation, y = female_lfpr)
) +
  geom_point(
    size = 2.5,
    color = point_colour
  ) +
  geom_smooth(
    method = "lm",
    se = TRUE
  ) +
  labs(
    title = "Female LFPR and Urbanisation",
    x = "Urban population (%)",
    y = "Female LFPR (%)"
  )
```

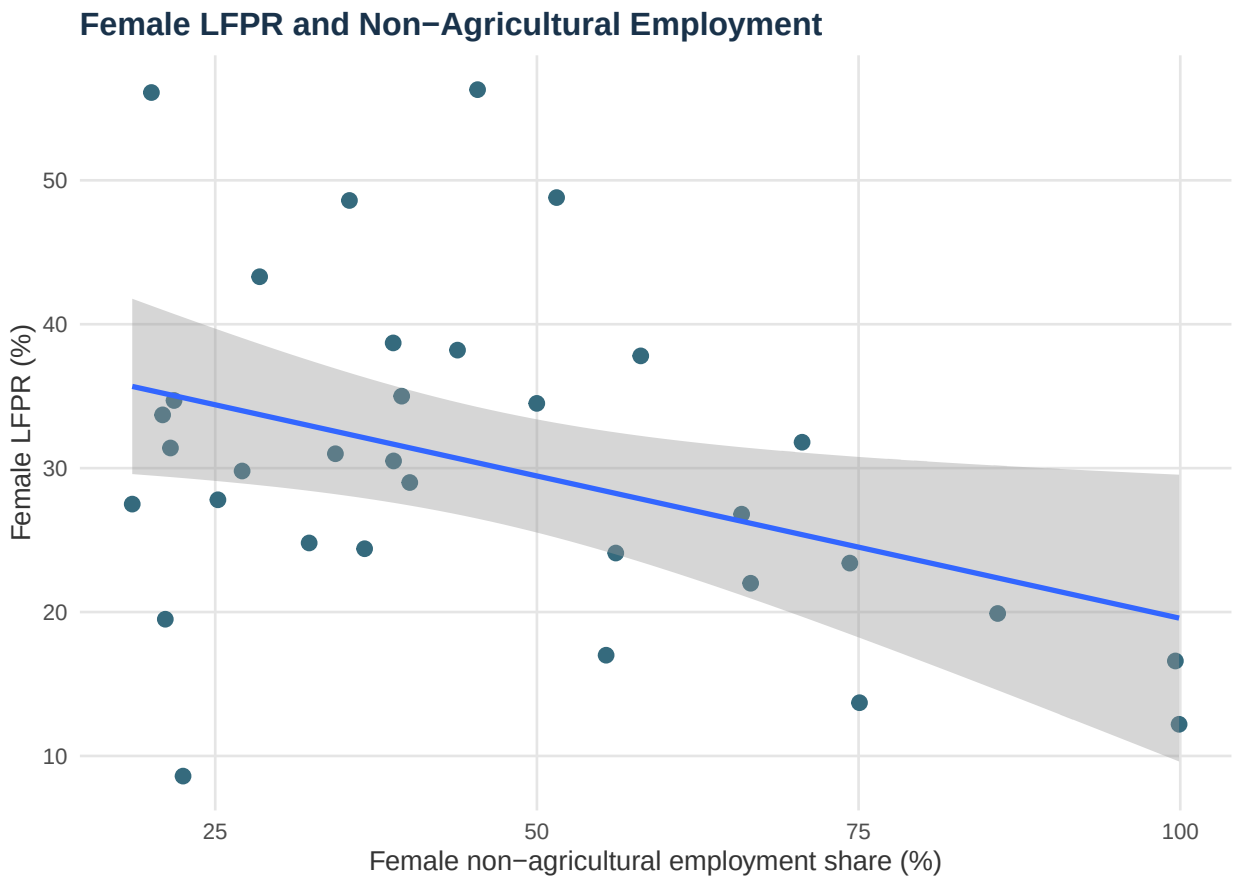
```
## `geom_smooth()` using formula = 'y ~ x'
```



5.4 Female LFPR and Non-Agricultural Employment

```
ggplot(
  df_analysis,
  aes(x = female_non_agri_share, y = female_lfpr)
) +
  geom_point(
    size = 2.5,
    color = point_colour
  ) +
  geom_smooth(
    method = "lm",
    se = TRUE
  ) +
  labs(
    title = "Female LFPR and Non-Agricultural Employment",
    x = "Female non-agricultural employment share (%)",
    y = "Female LFPR (%)"
  )
```

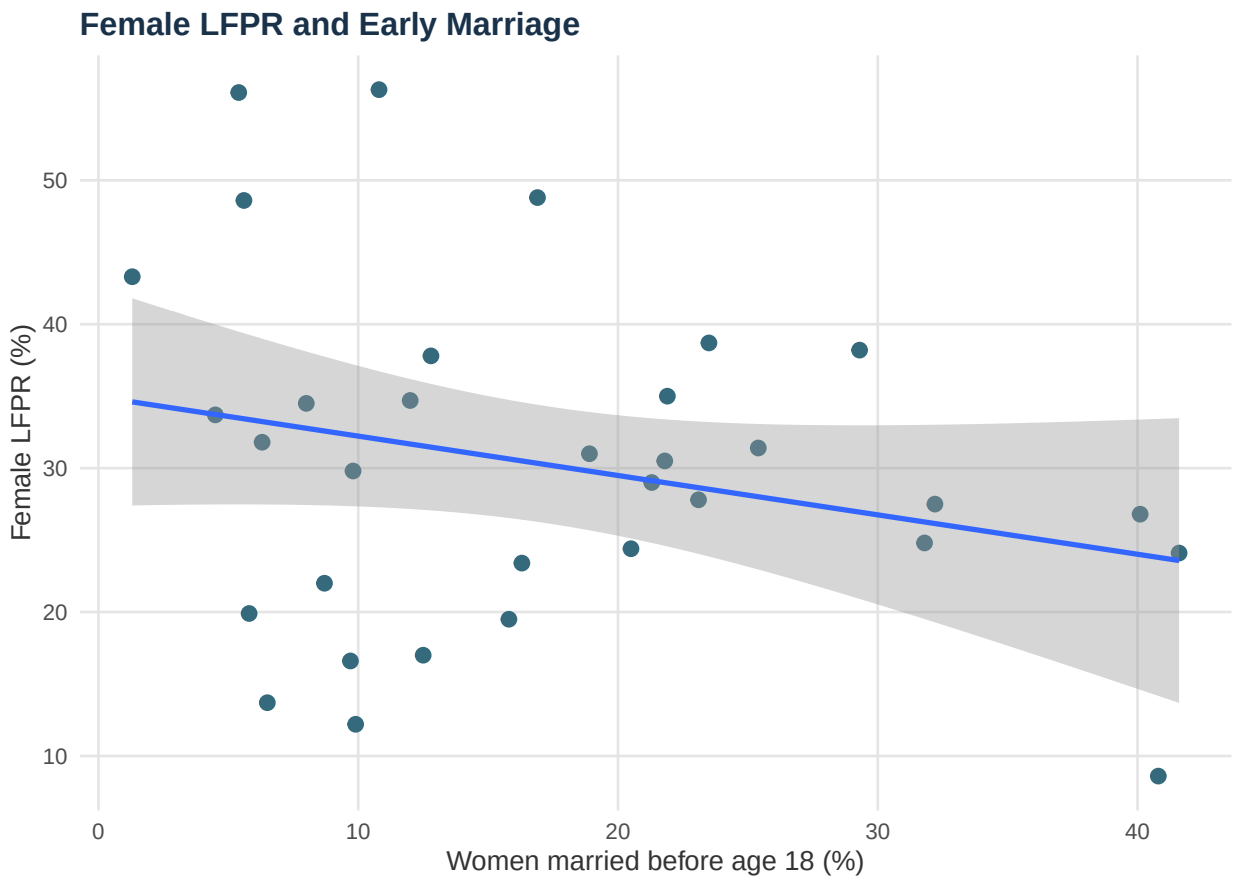
```
## `geom_smooth()` using formula = 'y ~ x'
```



5.5 Female LFPR and Early Marriage

```
ggplot(
  df_analysis,
  aes(x = married_before_18, y = female_lfpr)
) +
  geom_point(
    size = 2.5,
    color = point_colour
  ) +
  geom_smooth(
    method = "lm",
    se = TRUE
  ) +
  labs(
    title = "Female LFPR and Early Marriage",
    x = "Women married before age 18 (%)",
    y = "Female LFPR (%)"
  )
```

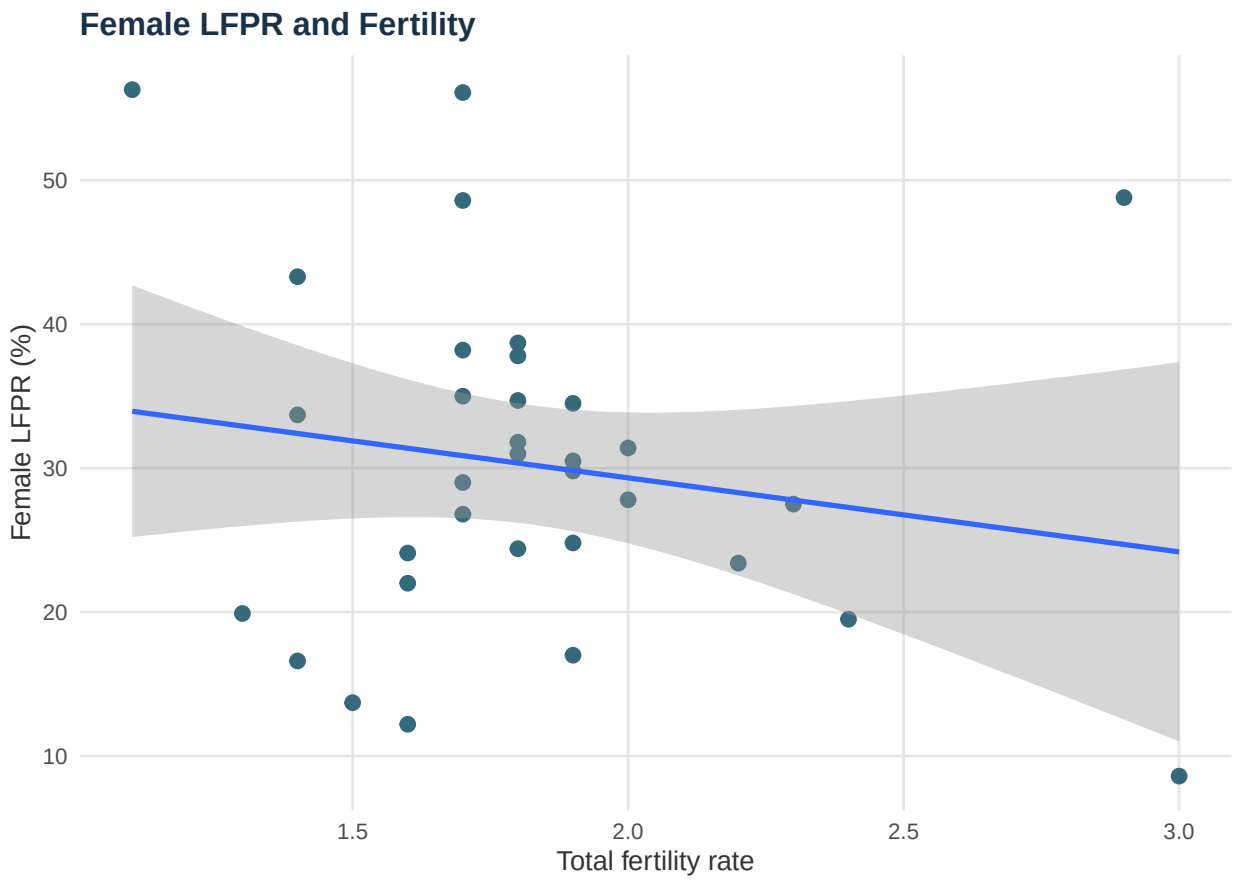
```
## `geom_smooth()` using formula = 'y ~ x'
```



5.6 Female LFPR and Fertility

```
ggplot(
  df_analysis,
  aes(x = tfr, y = female_lfpr)
) +
  geom_point(
    size = 2.5,
    color = point_colour
  ) +
  geom_smooth(
    method = "lm",
    se = TRUE
  ) +
  labs(
    title = "Female LFPR and Fertility",
    x = "Total fertility rate",
    y = "Female LFPR (%)"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6 Correlation Analysis

```
correlation_data <- df_analysis %>%
  select(
    female_lfpr,
    secondary_above,
    urbanisation,
    female_non_agri_share,
    married_before_18,
    tfr
  )

correlation_matrix <- cor(
  correlation_data,
  use = "pairwise.complete.obs"
)

round(correlation_matrix, 3)
```

```
##                       female_lfpr secondary_above urbanisation
```

```
## female_lfpr                 1.000          -0.115       -0.404
## secondary_above            -0.115           1.000        0.693
## urbanisation               -0.404           0.693        1.000
## female_non_agri_share      -0.391           0.614        0.808
## married_before_18          -0.261          -0.654       -0.348
## tfr                        -0.174          -0.453       -0.407
##                       female_non_agri_share married_before_18    tfr
## female_lfpr                          -0.391            -0.261 -0.174
## secondary_above                       0.614            -0.654 -0.453
## urbanisation                          0.808            -0.348 -0.407
## female_non_agri_share                 1.000            -0.223 -0.334
## married_before_18                    -0.223             1.000  0.432
## tfr                                  -0.334             0.432  1.000
```

```
# Correlation between female LFPR and log NSDP

cor(
  df_nsdp$female_lfpr,
  df_nsdp$log_nsdp,
  use = "complete.obs"
)
```

```
## [1] -0.2083093
```

```
df_analysis %>%
  filter(is.na(as.numeric(urbanisation))) %>%
  select(state_ut, urbanisation)
```

```
## # A tibble: 0 x 2
## # i 2 variables: state_ut <chr>, urbanisation <dbl>
```

For the NSDP correlation, the analysis uses the 32 states with available NSDP data.

```
cor(
  df_nsdp$female_lfpr,
  df_nsdp$log_nsdp,
  use = "complete.obs"
)
```

```
## [1] -0.2083093
```

7 Regression Analysis

7.1 Individual Models

Six separate ordinary least squares models are estimated to examine the association between female LFPR and each predictor individually.

```r
model_education <- lm(
  female_lfpr ~ secondary_above,
  data = df_analysis
)

model_nsdp <- lm(
  female_lfpr ~ log_nsdp,
  data = df_nsdp
)

model_urbanisation <- lm(
  female_lfpr ~ urbanisation,
  data = df_analysis
)

model_nonagri <- lm(
  female_lfpr ~ female_non_agri_share,
  data = df_analysis
)

model_marriage <- lm(
  female_lfpr ~ married_before_18,
  data = df_analysis
)

model_tfr <- lm(
  female_lfpr ~ tfr,
  data = df_analysis
)
```

7.2 Individual Model Results

```r
individual_models <- list(
  Education = model_education,
  Economic_Output = model_nsdp,
  Urbanisation = model_urbanisation,
  Non_Agricultural_Employment = model_nonagri,
  Early_Marriage = model_marriage,
  Fertility = model_tfr
)

individual_model_results <- bind_rows(
  lapply(
    names(individual_models),
    function(x) {
```

```
      tidy(individual_models[[x]]) %>%
        mutate(Model = x)
    }
  )
)

individual_model_results
```

```
## # A tibble: 12 x 6
##    term                   estimate std.error statistic  p.value Model
##    <chr>                     <dbl>     <dbl>     <dbl>    <dbl> <chr>
##  1 (Intercept)              34.9      7.51       4.64  5.98e- 5 Education
##  2 secondary_above          -0.127    0.197     -0.644 5.24e- 1 Education
##  3 (Intercept)              55.3     21.9        2.52  1.72e- 2 Economic_Output
##  4 log_nsdp                 -1.53     1.31      -1.17  2.53e- 1 Economic_Output
##  5 (Intercept)              38.3      3.80      10.1   2.58e-11 Urbanisation
##  6 urbanisation             -0.227    0.0922    -2.46  1.98e- 2 Urbanisation
##  7 (Intercept)              39.3      4.30       9.15  2.54e-10 Non_Agricultural~
##  8 female_non_agri_share    -0.198    0.0836    -2.37  2.44e- 2 Non_Agricultural~
##  9 (Intercept)              35.0      3.73       9.38  1.44e-10 Early_Marriage
## 10 married_before_18        -0.274    0.182     -1.50  1.43e- 1 Early_Marriage
## 11 (Intercept)              39.6      9.73       4.07  2.99e- 4 Fertility
## 12 tfr                      -5.14     5.21      -0.986 3.32e- 1 Fertility
```

7.3 Individual Model Comparison

```
individual_model_comparison <- data.frame(
  Model = names(individual_models),
  N = sapply(individual_models, nobs),
  R_squared = sapply(individual_models, function(x) summary(x)$r.squared),
  Adjusted_R_squared = sapply(
    individual_models,
    function(x) summary(x)$adj.r.squared
  )
)

individual_model_comparison
```

```
##                                                  Model  N  R_squared
## Education                                    Education 33 0.01320542
## Economic_Output                        Economic_Output 32 0.04339275
## Urbanisation                              Urbanisation 33 0.16311537
## Non_Agricultural_Employment Non_Agricultural_Employment 33 0.15292557
## Early_Marriage                        Early_Marriage 33 0.06795013
## Fertility                                  Fertility 33 0.03041427
```

```
##                              Adjusted_R_squared
## Education                        -0.0186266629
## Economic_Output                   0.0115058445
## Urbanisation                      0.1361190869
## Non_Agricultural_Employment       0.1256005933
## Early_Marriage                    0.0378840027
## Fertility                        -0.0008626906
```

```r
write.csv(
  individual_model_results,
  "individual_model_results.csv",
  row.names = FALSE
)

write.csv(
  individual_model_comparison,
  "individual_model_comparison.csv",
  row.names = FALSE
)
```

8 Combined Models

8.1 Education and Economic Output

This model examines education and economic output together.

Because NSDP is unavailable for two states, this model uses the 32 observations with complete NSDP data.

```r
model_education_economic <- lm(
  female_lfpr ~ secondary_above + log_nsdp,
  data = df_nsdp
)

summary(model_education_economic)
```

```
##
## Call:
## lm(formula = female_lfpr ~ secondary_above + log_nsdp, data = df_nsdp)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -22.055  -8.226  -1.200   7.239  27.361
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)      62.0572    23.5618   2.634   0.0134 *
## secondary_above  -0.1590     0.1955  -0.814   0.4225
## log_nsdp         -1.5851     1.3174  -1.203   0.2386
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.62 on 29 degrees of freedom
## Multiple R-squared:  0.06474,    Adjusted R-squared:  0.0002366
## F-statistic: 1.004 on 2 and 29 DF,  p-value: 0.3789
```

8.2 Education and Economic Output Diagnostics

```
vif(model_education_economic)
```

```
## secondary_above        log_nsdp
##        1.003074        1.003074
```

8.3 Demand-Side Factors

The demand-side model combines urbanisation and the female non-agricultural employment share.

```
model_demand <- lm(
  female_lfpr ~ urbanisation + female_non_agri_share,
  data = df_analysis
)

summary(model_demand)
```

```
##
## Call:
## lm(formula = female_lfpr ~ urbanisation + female_non_agri_share,
##     data = df_analysis)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -27.328  -5.726  -2.162   6.871  24.508
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            39.6575     4.3258   9.168 3.33e-10 ***
## urbanisation           -0.1421     0.1578  -0.901    0.375
## female_non_agri_share  -0.0944     0.1423  -0.664    0.512
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 10.95 on 30 degrees of freedom
## Multiple R-squared:  0.1752, Adjusted R-squared:  0.1202
## F-statistic: 3.187 on 2 and 30 DF,  p-value: 0.0556
```

8.4 Demand-Side Multicollinearity

```
vif(model_demand)
```

```
##          urbanisation female_non_agri_share
##              2.875535              2.875535
```

8.5 Supply-Side Factors

The supply-side model combines early marriage and fertility.

```
model_supply <- lm(
  female_lfpr ~ married_before_18 + tfr,
  data = df_analysis
)

summary(model_supply)
```

```
##
## Call:
## lm(formula = female_lfpr ~ married_before_18 + tfr, data = df_analysis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20.2986  -6.2237  -0.5381   5.5950  22.8962
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        38.4514     9.7192   3.956 0.000431 ***
## married_before_18  -0.2391     0.2046  -1.169 0.251641
## tfr                -2.2409     5.7463  -0.390 0.699317
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.61 on 30 degrees of freedom
## Multiple R-squared:  0.07265,    Adjusted R-squared:  0.01083
## F-statistic: 1.175 on 2 and 30 DF,  p-value: 0.3226
```

8.6 Supply-Side Multicollinearity

```
vif(model_supply)
```

```
## married_before_18               tfr
##          1.229166          1.229166
```

9 Full Model

The full model includes all six predictors simultaneously.

Because NSDP is included, the model is estimated using the 32 states with complete data on all six variables.

```
model_full <- lm(
  female_lfpr ~
    secondary_above +
    log_nsdp +
    urbanisation +
    female_non_agri_share +
    married_before_18 +
    tfr,
  data = df_nsdp
)

summary(model_full)
```

```
##
## Call:
## lm(formula = female_lfpr ~ secondary_above + log_nsdp + urbanisation +
##     female_non_agri_share + married_before_18 + tfr, data = df_nsdp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.6907  -7.3949   0.3639   6.1274  21.9887
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)           84.84312   24.17890   3.509  0.00173 **
## secondary_above        0.02316    0.32599   0.071  0.94393
## log_nsdp              -1.20531    1.41286  -0.853  0.40171
## urbanisation          -0.25259    0.16770  -1.506  0.14454
## female_non_agri_share -0.12256    0.15407  -0.796  0.43380
## married_before_18     -0.29957    0.26676  -1.123  0.27211
## tfr                   -8.48291    5.28246  -1.606  0.12086
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.985 on 25 degrees of freedom
## Multiple R-squared:  0.4047, Adjusted R-squared:  0.2619
## F-statistic: 2.833 on 6 and 25 DF,  p-value: 0.03035
```

9.1 Full Model Results

```
full_model_results <- tidy(model_full)

full_model_results
```

```
## # A tibble: 7 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)             84.8       24.2      3.51   0.00173
## 2 secondary_above          0.0232     0.326    0.0710 0.944
## 3 log_nsdp                -1.21       1.41    -0.853  0.402
## 4 urbanisation            -0.253      0.168   -1.51   0.145
## 5 female_non_agri_share   -0.123      0.154   -0.796  0.434
## 6 married_before_18       -0.300      0.267   -1.12   0.272
## 7 tfr                     -8.48       5.28    -1.61   0.121
```

```
write.csv(
  full_model_results,
  "full_model_results.csv",
  row.names = FALSE
)
```

10 Standardised Model

Standardising the predictors allows the relative size of coefficients to be compared across variables measured in different units.

```
df_standardised <- df_nsdp %>%
  mutate(
    female_lfpr_z = as.numeric(scale(female_lfpr)),
    secondary_above_z = as.numeric(scale(secondary_above)),
    log_nsdp_z = as.numeric(scale(log_nsdp)),
    urbanisation_z = as.numeric(scale(urbanisation)),
    female_non_agri_share_z = as.numeric(
      scale(female_non_agri_share)
    ),
```

```
    married_before_18_z = as.numeric(
      scale(married_before_18)
    ),
    tfr_z = as.numeric(scale(tfr))
  )

model_standardised <- lm(
  female_lfpr_z ~
    secondary_above_z +
    log_nsdp_z +
    urbanisation_z +
    female_non_agri_share_z +
    married_before_18_z +
    tfr_z,
  data = df_standardised
)

summary(model_standardised)
```

```
##
## Call:
## lm(formula = female_lfpr_z ~ secondary_above_z + log_nsdp_z +
##     urbanisation_z + female_non_agri_share_z + married_before_18_z +
##     tfr_z, data = df_standardised)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.17804 -0.63631  0.03131  0.52724  1.89204
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)              -1.851e-16  1.519e-01   0.000    1.000
## secondary_above_z         2.131e-02  3.000e-01   0.071    0.944
## log_nsdp_z               -1.646e-01  1.929e-01  -0.853    0.402
## urbanisation_z           -4.566e-01  3.031e-01  -1.506    0.145
## female_non_agri_share_z -2.449e-01  3.079e-01  -0.796    0.434
## married_before_18_z      -2.815e-01  2.507e-01  -1.123    0.272
## tfr_z                    -2.883e-01  1.795e-01  -1.606    0.121
##
## Residual standard error: 0.8591 on 25 degrees of freedom
## Multiple R-squared:  0.4047, Adjusted R-squared:  0.2619
## F-statistic: 2.833 on 6 and 25 DF,  p-value: 0.03035
```

10.1 Standardised Model Results

```
standardised_model_results <- tidy(model_standardised)

standardised_model_results
```

```
## # A tibble: 7 x 5
##   term                     estimate std.error statistic p.value
##   <chr>                       <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)             -1.85e-16     0.152 -1.22e-15   1.000
## 2 secondary_above_z        2.13e- 2     0.300  7.10e- 2   0.944
## 3 log_nsdp_z              -1.65e- 1     0.193 -8.53e- 1   0.402
## 4 urbanisation_z          -4.57e- 1     0.303 -1.51e+ 0   0.145
## 5 female_non_agri_share_z -2.45e- 1     0.308 -7.96e- 1   0.434
## 6 married_before_18_z     -2.82e- 1     0.251 -1.12e+ 0   0.272
## 7 tfr_z                   -2.88e- 1     0.180 -1.61e+ 0   0.121
```

```
write.csv(
  standardised_model_results,
  "standardised_model_results.csv",
  row.names = FALSE
)
```

11 Model Diagnostics

11.1 Multicollinearity

Variance inflation factors are calculated for the full model.

```
full_model_vif <- vif(model_full)

full_model_vif
```

```
##       secondary_above              log_nsdp          urbanisation
##              3.778753              1.562677              3.859079
## female_non_agri_share     married_before_18                   tfr
##              3.981113              2.639105              1.353351
```

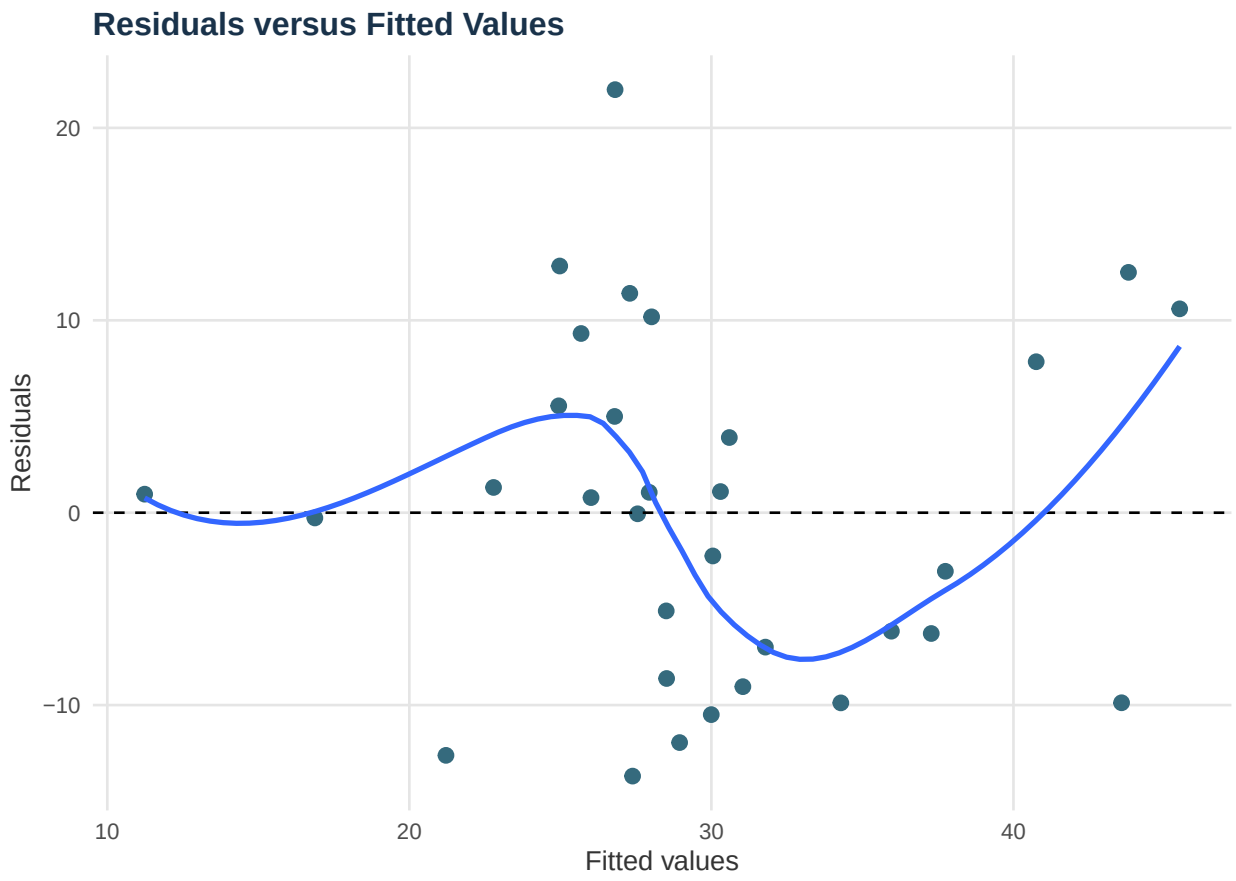
11.2 Regression Diagnostics

The following diagnostics are used to assess whether the linear regression assumptions raise substantial concerns.

11.3 Residuals versus Fitted Values

```
ggplot(
  data.frame(
    fitted = fitted(model_full),
    residuals = residuals(model_full)
  ),
  aes(x = fitted, y = residuals)
) +
  geom_point(
    size = 2.5,
    color = point_colour
  ) +
  geom_hline(
    yintercept = 0,
    linetype = "dashed"
  ) +
  geom_smooth(
    method = "loess",
    se = FALSE
  ) +
  labs(
    title = "Residuals versus Fitted Values",
    x = "Fitted values",
    y = "Residuals"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

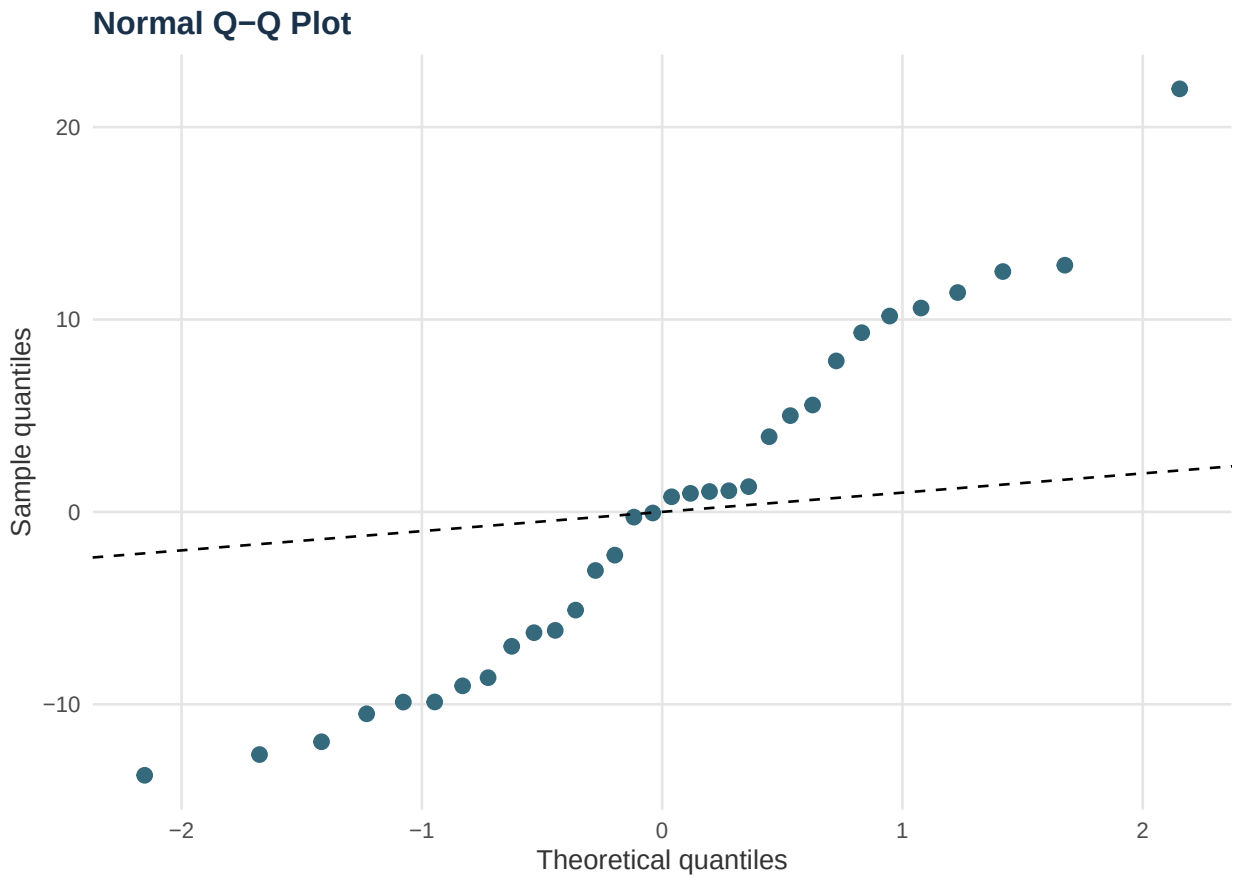


11.4 Normal Q-Q Plot

```
qq_data <- data.frame(
  theoretical = qqnorm(
    residuals(model_full),
    plot.it = FALSE
)$x,
  sample = qqnorm(
    residuals(model_full),
    plot.it = FALSE
)$y
)

ggplot(
  qq_data,
  aes(x = theoretical, y = sample)
) +
  geom_point(
    size = 2.5,
    color = point_colour
```

```
  ) +
  geom_abline(
    intercept = 0,
    slope = 1,
    linetype = "dashed"
  ) +
  labs(
    title = "Normal Q-Q Plot",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  )
```



11.5 Influential Observations

Cook's distance is used to identify observations that may exert disproportionate influence on the full model.

```
cooks_values <- cooks.distance(model_full)

cooks_data <- data.frame(
  state_ut = df_nsdp$state_ut,
```

```
  cooks_distance = cooks_values
) %>%
  arrange(desc(cooks_distance))

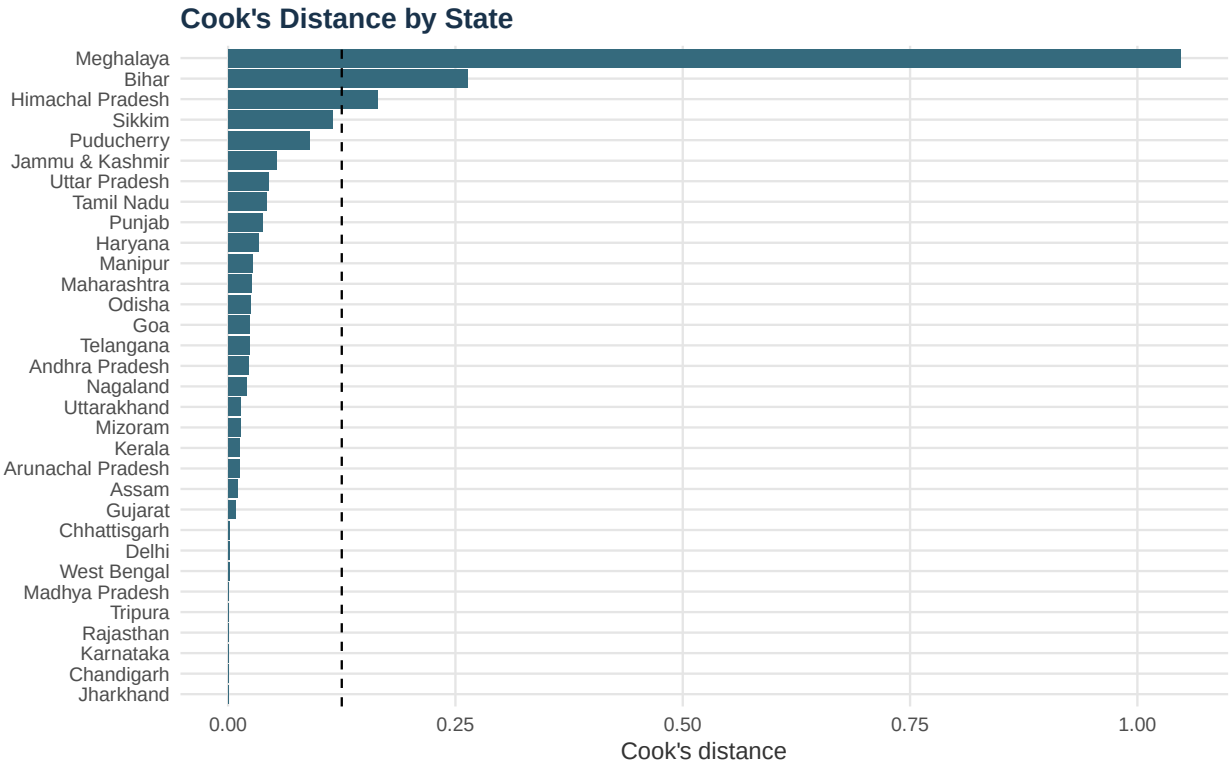
cooks_data
```

```
##             state_ut cooks_distance
## 19         Meghalaya   1.047414e+00
## 4              Bihar   2.630670e-01
## 11  Himachal Pradesh   1.643096e-01
## 26            Sikkim   1.144594e-01
## 23        Puducherry   8.977332e-02
## 12   Jammu & Kashmir   5.370909e-02
## 30     Uttar Pradesh   4.441209e-02
## 27        Tamil Nadu   4.213001e-02
## 24            Punjab   3.824955e-02
## 10           Haryana   3.350131e-02
## 18           Manipur   2.661092e-02
## 17       Maharashtra   2.550139e-02
## 22            Odisha   2.431549e-02
## 8                Goa   2.389893e-02
## 28         Telangana   2.345387e-02
## 1     Andhra Pradesh   2.302628e-02
## 21          Nagaland   1.997214e-02
## 31       Uttarakhand   1.362801e-02
## 20           Mizoram   1.334501e-02
## 15            Kerala   1.260966e-02
## 2  Arunachal Pradesh   1.251948e-02
## 3              Assam   1.074296e-02
## 9            Gujarat   8.660003e-03
## 6       Chhattisgarh   1.649503e-03
## 7              Delhi   1.483212e-03
## 32       West Bengal   1.454167e-03
## 16    Madhya Pradesh   1.101803e-03
## 29           Tripura   9.371998e-04
## 25         Rajasthan   2.108621e-04
## 14         Karnataka   1.885236e-04
## 5          Chandigarh   1.523981e-04
## 13         Jharkhand   1.434846e-06
```

```
write.csv(
  cooks_data,
  "cooks_distance.csv",
  row.names = FALSE
)
```

11.6 Cook's Distance Plot

```
ggplot(
  cooks_data,
  aes(
    x = reorder(state_ut, cooks_distance),
    y = cooks_distance
  )
) +
  geom_col(
    fill = point_colour
  ) +
  geom_hline(
    yintercept = 4 / nrow(df_nsdp),
    linetype = "dashed"
  ) +
  coord_flip() +
  labs(
    title = "Cook's Distance by State",
    x = NULL,
    y = "Cook's distance"
  )
```



12 Model Comparison

The models are compared using R-squared and adjusted R-squared.

```
model_comparison <- data.frame(
  Model = c(
    "Education",
    "Economic output",
    "Urbanisation",
    "Non-agricultural employment",
    "Early marriage",
    "Fertility",
    "Education + economic output",
    "Demand-side factors",
    "Supply-side factors",
    "Full model"
  ),
  N = c(
    nobs(model_education),
    nobs(model_nsdp),
    nobs(model_urbanisation),
    nobs(model_nonagri),
    nobs(model_marriage),
    nobs(model_tfr),
    nobs(model_education_economic),
    nobs(model_demand),
    nobs(model_supply),
    nobs(model_full)
  ),
  R_squared = c(
    summary(model_education)$r.squared,
    summary(model_nsdp)$r.squared,
    summary(model_urbanisation)$r.squared,
    summary(model_nonagri)$r.squared,
    summary(model_marriage)$r.squared,
    summary(model_tfr)$r.squared,
    summary(model_education_economic)$r.squared,
    summary(model_demand)$r.squared,
    summary(model_supply)$r.squared,
    summary(model_full)$r.squared
  ),
  Adjusted_R_squared = c(
    summary(model_education)$adj.r.squared,
    summary(model_nsdp)$adj.r.squared,
    summary(model_urbanisation)$adj.r.squared,
    summary(model_nonagri)$adj.r.squared,
    summary(model_marriage)$adj.r.squared,
```

```
    summary(model_tfr)$adj.r.squared,
    summary(model_education_economic)$adj.r.squared,
    summary(model_demand)$adj.r.squared,
    summary(model_supply)$adj.r.squared,
    summary(model_full)$adj.r.squared
  )
)

model_comparison
```

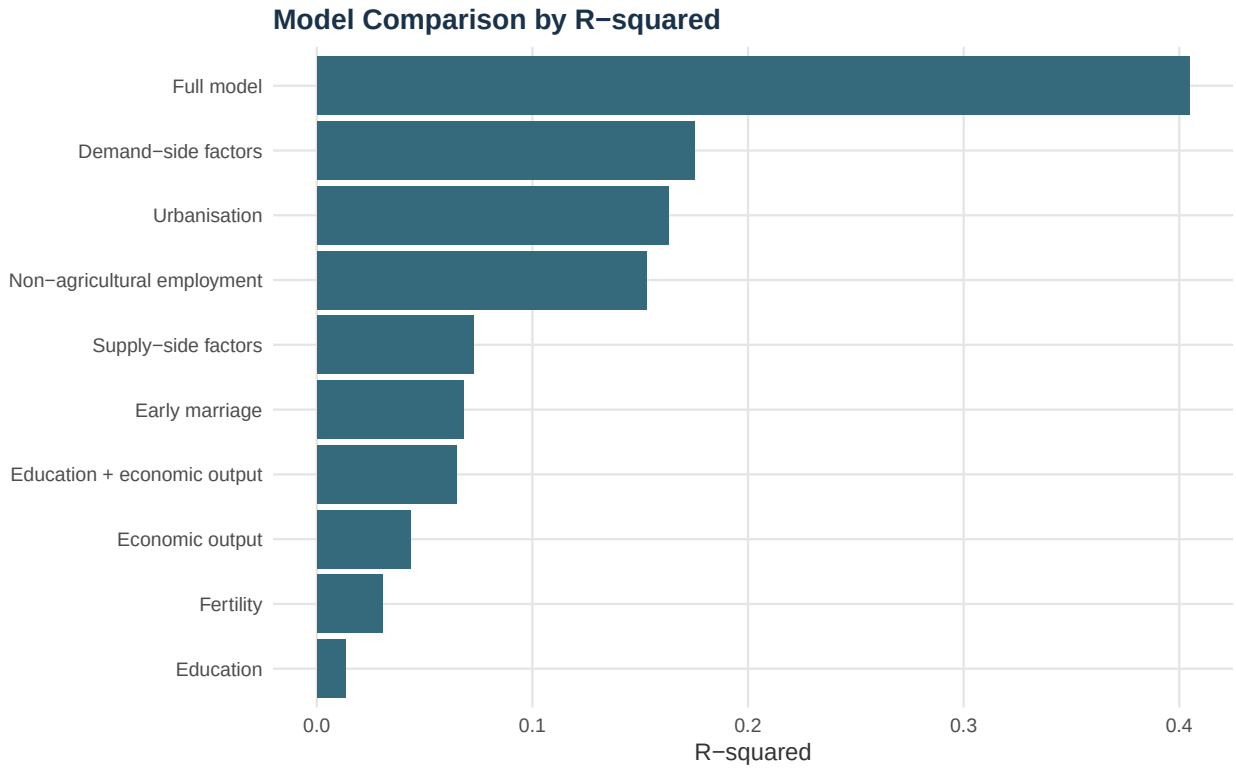
```
##                          Model  N  R_squared Adjusted_R_squared
## 1                    Education 33 0.01320542      -0.0186266629
## 2              Economic output 32 0.04339275       0.0115058445
## 3                 Urbanisation 33 0.16311537       0.1361190869
## 4  Non-agricultural employment 33 0.15292557       0.1256005933
## 5               Early marriage 33 0.06795013       0.0378840027
## 6                    Fertility 33 0.03041427      -0.0008626906
## 7  Education + economic output 32 0.06473748       0.0002366135
## 8          Demand-side factors 33 0.17522042       0.1202351181
## 9          Supply-side factors 33 0.07265095       0.0108276822
## 10                  Full model 32 0.40474754       0.2618869440
```

```
write.csv(
  model_comparison,
  "model_comparison.csv",
  row.names = FALSE
)
```

12.1 R-squared Comparison

```
ggplot(
  model_comparison,
  aes(
    x = reorder(Model, R_squared),
    y = R_squared
  )
) +
  geom_col(
    fill = point_colour
  ) +
  coord_flip() +
  labs(
    title = "Model Comparison by R-squared",
    x = NULL,
```

```
    y = "R-squared"
  )
```



13 Findings

The state-level analysis examines the association between female labour force participation and six structural and social indicators: female secondary education, economic output, urbanisation, female non-agricultural employment, early marriage and fertility. The individual models provide an initial comparison of the strength of these associations. The combined models then examine whether these relationships remain when related predictors are considered together. The results should be interpreted as **associations rather than causal effects**. The cross-sectional state-level design does not establish whether changes in any individual predictor would directly cause changes in female labour force participation.

```
# Extract key results for reporting

key_results <- data.frame(
  Model = c(
    "Education",
    "Economic output",
    "Urbanisation",
    "Non-agricultural employment",
    "Early marriage",
```

```
    "Fertility",
    "Full model"
  ),
  N = c(
    nobs(model_education),
    nobs(model_nsdp),
    nobs(model_urbanisation),
    nobs(model_nonagri),
    nobs(model_marriage),
    nobs(model_tfr),
    nobs(model_full)
  ),
  R_squared = c(
    summary(model_education)$r.squared,
    summary(model_nsdp)$r.squared,
    summary(model_urbanisation)$r.squared,
    summary(model_nonagri)$r.squared,
    summary(model_marriage)$r.squared,
    summary(model_tfr)$r.squared,
    summary(model_full)$r.squared
  )
)

key_results
```

```
##                         Model  N  R_squared
## 1                   Education 33 0.01320542
## 2             Economic output 32 0.04339275
## 3                Urbanisation 33 0.16311537
## 4 Non-agricultural employment 33 0.15292557
## 5               Early marriage 33 0.06795013
## 6                    Fertility 33 0.03041427
## 7                  Full model 32 0.40474754
```

14 Limitations

This analysis has several limitations.

First, the analysis is conducted at the state/UT level and therefore does not capture differences between individuals, households, districts or communities within states.

Second, the analysis is cross-sectional and uses indicators from different years where the most appropriate publicly available measures differ in reference period. The results therefore describe patterns of association rather than temporal or causal relationships.

Third, the sample is relatively small, with 34 states/UTs in the general analytical sample and 32 observations available for models involving NSDP. Estimates from the multivariable models should therefore be interpreted cautiously.

Fourth, state-level indicators may conceal substantial heterogeneity within states. A state with a relatively high average female LFPR, for example, may still contain districts or groups with very low participation.

Finally, the analysis is limited to the indicators that could be obtained and harmonised consistently across states. Other potentially important factors—including wages, childcare availability, safety, occupational segregation, household income, social norms and the availability of suitable employment—are not directly captured in the model.

These limitations mean that the results are best viewed as identifying **state-level patterns that merit further investigation**, rather than as estimates of causal effects.

15 Reproducibility

All data preparation, transformations, statistical models, diagnostics and visualisations in this analysis were conducted in R.

The principal packages used were:

```
package_list <- c(
  "readxl",
  "dplyr",
  "ggplot2",
  "broom",
  "car",
  "tidyr",
  "knitr",
  "kableExtra"
)

package_list
```

```
## [1] "readxl"     "dplyr"      "ggplot2"    "broom"      "car"
## [6] "tidyr"      "knitr"      "kableExtra"
```

15.1 Exported Results

The analysis exports the following files to the project directory:

```
exported_files <- c(
  "individual_model_results.csv",
  "individual_model_comparison.csv",
  "full_model_results.csv",
  "standardised_model_results.csv",
  "model_comparison.csv",
  "cooks_distance.csv"
)
```

```
exported_files
```

```
## [1] "individual_model_results.csv"    "individual_model_comparison.csv"
## [3] "full_model_results.csv"          "standardised_model_results.csv"
## [5] "model_comparison.csv"            "cooks_distance.csv"
```

16 R Session Information

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 10 x64 (build 19045)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_India.utf8  LC_CTYPE=English_India.utf8
## [3] LC_MONETARY=English_India.utf8 LC_NUMERIC=C
## [5] LC_TIME=English_India.utf8
##
## time zone: Asia/Calcutta
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] kableExtra_1.4.1 knitr_1.51       tidyr_1.3.2      car_3.1-5
## [5] carData_3.0-6    broom_1.0.13     ggplot2_4.0.3    dplyr_1.2.1
## [9] readxl_1.5.0
##
## loaded via a namespace (and not attached):
##  [1] utf8_1.2.6         generics_0.1.4     xml2_1.6.0         lattice_0.22-9
##  [5] stringi_1.8.9      digest_0.6.39      magrittr_2.0.5     evaluate_1.0.5
##  [9] grid_4.6.1         RColorBrewer_1.1-3 fastmap_1.2.0      Matrix_1.7-5
## [13] cellranger_1.1.0   backports_1.5.1    Formula_1.2-6      mgcv_1.9-4
## [17] purrr_1.2.2        viridisLite_0.4.3  scales_1.4.0       textshaping_1.0.5
## [21] abind_1.4-8        cli_3.6.6          rlang_1.3.0        splines_4.6.1
## [25] withr_3.0.3        yaml_2.3.12        otel_0.2.0         tools_4.6.1
## [29] vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5    stringr_1.6.0
## [33] pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6       glue_1.8.1
## [37] systemfonts_1.3.2  xfun_0.60          tibble_3.3.1       tidyselect_1.2.1
```

```
## [41] rstudioapi_0.19.0  farver_2.1.2       nlme_3.1-169       htmltools_0.5.9
## [45] rmarkdown_2.32     svglite_2.2.2      labeling_0.4.3     compiler_4.6.1
## [49] S7_0.2.2
```